

Diarrhea in Adults

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Diarrhea is an increase in the volume, wateriness, or frequency of bowel movements.

The frequency of bowel movements alone is not the defining feature of diarrhea. Some people normally move their bowels 3 to 5 times a day. People who eat large amounts of vegetable fiber may produce more than a pound (1/2 kilogram) of stool a day, but the stool in such cases is well formed and not watery.

Diarrhea is often accompanied by gas, cramping, an urgency to defecate, and, if the diarrhea is caused by an infectious organism or a toxic substance, nausea and vomiting.

(See also [Diarrhea in Children](#).)

Complications

Diarrhea can lead to [dehydration](#) and a loss of [electrolytes](#), such as sodium, potassium, magnesium, chloride, and bicarbonate, from the blood. If large amounts of fluid and electrolytes are lost, the person feels weak, and blood pressure can drop enough to cause [fainting](#) (syncope), [heart rhythm abnormalities](#) (arrhythmias), and other serious disorders. At particular risk are people who are very young, very old, or weakened and people with very severe diarrhea.

Causes of Diarrhea

There are many different causes of diarrhea, depending on how long the diarrhea has lasted (see table [Some Causes and Features of Diarrhea](#)).

The **most common causes of acute diarrhea** (lasting less than 4 days) are

- Infection with viruses, bacteria, or parasites ([gastroenteritis](#))

- [Food poisoning](#)
- Side effects of a medication

The **most common causes of chronic diarrhea** (lasting more than 4 weeks) are

- [Irritable bowel syndrome](#)
- [Inflammatory bowel disease](#)
- Side effects of a medication
- [Malabsorption](#)

Diarrhea that has been present for more than 4 weeks may be a lingering case of acute diarrhea or the early stage of a disorder that causes chronic diarrhea.

Classification

Normally, stool is 65 to 95% water. Diarrhea occurs when not enough water is removed from the stool, making the stool loose and poorly formed. Stool may contain too much water if it:

- Passes too quickly through the digestive tract
- Contains certain substances that prevent the large intestine from absorbing water
- Contains excess water secreted by the intestines

Rapid passage (transit) of stool is a common cause of diarrhea. For stool to have normal consistency, it must remain in the large intestine for a certain amount of time. Stool that leaves the large intestine too quickly is watery.

Many medical conditions and treatments can decrease the amount of time that stool stays in the large intestine:

- An overactive thyroid ([hyperthyroidism](#))
- [Zollinger-Ellison syndrome](#) (a condition of over-production of acid by a tumor)
- Surgical removal of part of the stomach, small intestine, or large intestine or of the gallbladder ([cholecystectomy](#))
- Surgical bypass of part of the intestine
- Inflammatory bowel disease (such as [ulcerative colitis](#))
- Medications such as antacids containing magnesium, laxatives, prostaglandins, serotonin, and even caffeine

Many foods, especially those that are acidic or have a very high amount of sugar (such as waffle or maple syrup), can increase the rate of transit (see table

Foods and Beverages That May Cause Diarrhea Some people are intolerant of specific foods and always develop diarrhea after eating them.

Stress and anxiety are also common causes.

Osmotic diarrhea occurs when certain substances that cannot be absorbed through the colon wall remain in the intestine. These substances cause excessive amounts of water to remain in the stool, leading to diarrhea.

Certain foods (such as some fruits and beans) and sugar substitutes in dietetic foods, candy, and chewing gum (for example, hexitols, sorbitol, and mannitol) can cause osmotic diarrhea.

Lactase deficiency also can lead to osmotic diarrhea. Lactase is an enzyme normally found in the small intestine that converts lactose (milk sugar) to glucose and galactose, so that it can be absorbed into the bloodstream. When people with lactase deficiency drink milk or eat dairy products, lactose is not digested. As lactose accumulates in the intestine, it causes osmotic diarrhea—a condition known as lactose intolerance. The severity of osmotic diarrhea depends on how much of the osmotic substance is consumed. Diarrhea stops soon after the person stops eating or drinking the substance.

Blood in the digestive tract also acts as an osmotic agent and results in black, tarry stools (melena).

Another cause of osmotic diarrhea is an overgrowth of normal intestinal bacteria or the growth of bacteria normally not found in the intestines.

Antibiotics can cause osmotic diarrhea by destroying the normal intestinal bacteria (see, for example, *Clostridioides difficile*-induced colitis).

Secretory diarrhea occurs when the small and large intestines secrete salts (especially sodium chloride) and water into the stool.

Certain toxins—such as the toxin produced by a cholera infection or during some viral infections—can cause these secretions. Infections by certain bacteria (for example, *Campylobacter*) and parasites (for example, *Cryptosporidium*) can also stimulate secretions. The diarrhea can be massive—more than 1 quart (1 liter) of stool an hour in cholera.

Other substances that cause salt and water secretion include certain laxatives, such as castor oil, and bile acids (which may build up after surgery to remove part of the small intestine).

Certain rare tumors, such as [carcinoid tumors](#), [gastrinomas](#), and [VIPomas](#), also can cause secretory diarrhea, as can some [polyps](#).

Inflammatory diarrhea occurs when the lining of the large intestine becomes inflamed, ulcerated, or engorged and releases proteins, blood, mucus, and other fluids, which increase the bulk and fluid content of the stool. This type of diarrhea can be caused by many diseases, including [ulcerative colitis](#), [Crohn disease](#), tuberculosis, and cancers such as [lymphoma](#) and [adenocarcinoma](#). When the lining of the rectum is affected, people often feel an urgent need to move their bowels and have frequent bowel movements because the inflamed rectum is more sensitive to expansion (distention) by stool.

Diarrhea caused by malabsorption is characterized by oil or grease in the stool and an oily rim around the toilet bowl after stool is flushed. Malabsorption of bile salts, which can result from certain disorders, can cause diarrhea by stimulating water and electrolyte secretion. The stools have a green or orange color.



Foods and Beverages That May Cause Diarrhea

Food or Beverage	Ingredient Causing Diarrhea
Sugar-free gum, mints, sweet cherries, or prunes	Hexitols, sorbitol, or mannitol
Apple juice, pear juice, grapes, honey, dates, nuts, figs, soft drinks (especially fruit flavors), prunes, or waffle or maple syrup	Fructose
Milk, ice cream, yogurt, or soft cheese	Lactose
Coffee, tea, cola drinks, or some over-the-counter headache remedies	Caffeine

